

Autoinjection: A New Class of Mathematical Mapping and Its Formal Properties

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Abstract

This paper introduces autoinjection as a formally distinct class of mathematical mapping not present in the classical taxonomy established by the Bourbaki group. Classical set theory recognizes three classes of functions — injection, surjection, and bijection — based on how elements of a domain correspond to elements of a codomain treated as external to the domain. Autoinjection is defined as a mapping in which the codomain is a subset of the domain itself, such that the output of a function is an element of its own input set. This property, which classical mathematics treats as paradoxical or pathological, is here proposed as a fundamental and well-defined structural class. The formal definition, basic properties, and relationship to existing mathematical structures are presented. The implications for self-referential systems, recursive structures, and the foundations of set theory are discussed.

Keywords: autoinjection, self-reference, function classification, set theory, Bourbaki, recursive mapping, foundations of mathematics

1. Introduction

The classical taxonomy of functions, formalized by the Bourbaki group in the mid-twentieth century, defines three fundamental classes based on the relationship between domain and codomain (Bourbaki, 1968). An injective function maps distinct elements of the domain to distinct elements of the codomain. A surjective function maps the domain onto the entire codomain. A bijective function satisfies both conditions simultaneously. In all three cases, domain and codomain are treated as distinct sets, or at minimum as sets whose relationship is defined externally.

This taxonomy is powerful and has served as the foundation of modern mathematics. However, it contains a structural gap. It does not formally address the case in which the output of a function is an element of the input domain itself — that is, the case in which

codomain $\subseteq$ domain at the level of the function's output. This case is typically either excluded by convention, assigned to the separate study of fixed points (Brouwer, 1911; Tarski, 1955), or treated as problematic through association with self-referential paradoxes such as Russell's paradox (Russell, 1903).

This paper proposes that this gap represents a missing class of mapping that deserves formal recognition on its own terms. We call this class autoinjection. Unlike the study of fixed points, which concerns the specific condition $f(x) = x$, autoinjection concerns the broader structural condition that the range of f is contained within its domain. Unlike the avoidance strategies of naive and axiomatic set theory (Zermelo, 1908), autoinjection does not treat self-containment as a problem to be eliminated. It treats self-containment as a primitive structural property to be defined and studied.

2. Formal Definition

Let A be a non-empty set. Let f be a function defined on A .

Definition 1 (Autoinjective Function). A function $f : A \rightarrow A$ is autoinjective if and only if the range of f is a subset of the domain of f . That is:

$$f : A \rightarrow A \text{ such that } \text{range}(f) \subseteq A$$

This condition is satisfied trivially for any function $f : A \rightarrow A$ in the sense that the codomain is declared to be A . The substantive content of autoinjection, however, concerns the case in which A is defined as a set that contains itself as an element — that is, where $A \in A$.

Definition 2 (Autoinjective Set). A set A is autoinjective if $A \in A$.

This definition directly engages the condition that classical set theory, in both naive and axiomatic forms, has sought to avoid. In Zermelo-Fraenkel set theory, the axiom of regularity (also called the axiom of foundation) explicitly prohibits $A \in A$ (Zermelo, 1908; von Neumann, 1925). Autoinjection proposes that rather than prohibiting this condition, we may define a class of structures for which it holds and study their properties formally.

Definition 3 (Autoinjective Mapping). A mapping f is an autoinjective mapping if it satisfies the following: there exists a set E such that $E \in E$, and $f : E \rightarrow E$ such that for some element $e \in E$, $f(e) = E$.

In plain terms: the function maps some element of the set to the set itself, which is also a member of itself. The output is the container. The part maps to the whole, and the whole is a part of itself.

3. Relationship to Classical Mapping Classes

The three classical classes of Bourbaki can be restated in a unified form as follows. Let $f : X \rightarrow Y$. The classical classes concern the relationship between X and Y as distinct sets. Injection requires that f preserves distinctness. Surjection requires that f covers Y entirely. Bijection requires both.

Autoinjection introduces a fourth structural condition that is orthogonal to these three. It does not concern how elements of X correspond to elements of Y . It concerns the relationship between X and Y themselves — specifically, the condition $Y \subseteq X$ and $X \in X$.

This can be represented schematically:

Class	Condition
Injection	$f(a) = f(b) \Rightarrow a = b$
Surjection	$\forall y \in Y, \exists x \in X : f(x) = y$
Bijection	Injection $\wedge$ Surjection
Autoinjection	$\text{range}(f) \subseteq \text{domain}(f), \text{domain}(f) \in \text{domain}(f)$

Autoinjection is not a special case of any of the three classical classes, nor is it their negation. It is a structurally distinct class defined by the self-containment of the domain rather than by the correspondence properties of the mapping.

4. Relationship to Fixed Point Theory

Fixed point theory, developed by Brouwer (1911) and extended by Tarski (1955) and others, concerns the condition $f(x) = x$ — that is, the existence of elements that the function maps to themselves. This is a point-level condition. Autoinjection is a set-level condition. It concerns not whether any particular element is preserved by the mapping, but whether the set itself is an element of itself and whether the mapping's range falls within that set.

Every fixed point of a function $f : A \rightarrow A$ satisfies $f(x) = x \in A$, which is consistent with autoinjection. But autoinjection does not require fixed points. A function may be autoinjective without any element satisfying $f(x) = x$. The two concepts are therefore distinct, though compatible.

5. Relationship to Self-Reference and Paradox

Russell's paradox arises from the assumption that for any property P , there exists a set of all objects with property P (Russell, 1903). The set $R = \{x : x \notin x\}$ leads to contradiction. The

classical response, formalized by Zermelo (1908), is to restrict set formation so that $A \in A$ is prohibited.

Autoinjection takes a different approach. Rather than prohibiting self-membership, it proposes that self-membership is a definable property belonging to a specific class of sets. The contradiction in Russell's paradox arises not from self-membership per se but from unrestricted comprehension — the assumption that any property defines a set. Autoinjective set theory does not require unrestricted comprehension. It requires only that we admit the existence of sets A for which $A \in A$, defined by explicit construction rather than by unrestricted abstraction.

This approach has precedent in non-well-founded set theory, developed independently by Aczel (1988), which replaces the axiom of foundation with an anti-foundation axiom permitting circular membership. Autoinjection arrives at a compatible structural insight through a different route — from the observation that the classical mapping taxonomy is incomplete — and formalizes it as a property of functions rather than primarily as a property of sets.

6. A Canonical Example: The Set of Everything

Consider the set E defined informally as the set of all things — every object, concept, relation, and set, including E itself. This set satisfies $E \in E$ by definition: E is a thing, and E contains all things, therefore E contains itself.

A function $f : E \rightarrow E$ defined by $f(x) = E$ for all $x \in E$ is autoinjective in the sense of Definition 3. Every element of E is mapped to E , which is itself an element of E . The output of the function is always a member of the domain. The range equals $\{E\}$, which is a subset of E .

This example illustrates the key structural property of autoinjection: the whole is an element of itself, and the function can map any part to the whole. The part-whole relationship is not hierarchical but self-containing. Classical mathematics assumes strict hierarchy between domain and codomain. Autoinjection replaces hierarchy with self-containment as the fundamental structural relationship.

7. Formal Properties of Autoinjective Mappings

We state the following properties without full proof, noting that formal proofs require a foundational framework that either adopts an anti-foundation axiom (Aczel, 1988) or explicitly defines the class of autoinjective sets as a primitive.

Property 1 (Closure). If $f : A \rightarrow A$ is autoinjective and $A \in A$, then for any $g : A \rightarrow A$ with $\text{range}(g) \subseteq \text{range}(f)$, g is also autoinjective.

Property 2 (Composition). If f and g are autoinjective mappings on A , the composition $g \circ f$ is autoinjective on A .

Property 3 (Recursion). An autoinjective mapping $f : A \rightarrow A$ admits a natural recursion: define $f^0 = \text{identity}$, $f^n = f \circ f^{n-1}$. Since $\text{range}(f^n) \subseteq A$ for all $n \in \mathbb{N}$, every iterate of f is also autoinjective on A .

Property 4 (Self-evaluation). Because $A \in A$, the function f can receive itself as input: $f(A)$ is defined and equals some element of A . This permits the mapping to evaluate its own domain.

Property 4 is the formal basis for the application of autoinjection to self-improving and self-evaluating systems, including recursive cognitive architectures and artificial intelligence systems that must evaluate their own outputs (Voronin, 2021).

8. Discussion

The formal gap in the Bourbaki mapping taxonomy is narrow but consequential. By defining domain and codomain as always external to one another — or by treating self-containment as a degenerate case — classical mathematics has systematically excluded a class of structures that appear naturally in recursive systems, self-referential reasoning, and any formal treatment of totality.

Autoinjection fills this gap by providing a fourth class of mapping defined not by correspondence properties internal to the function but by the self-containment property of the domain. It does not replace the three classical classes. It extends the taxonomy to include the case that classical mathematics, for foundational reasons, has preferred to avoid.

The relationship to non-well-founded set theory (Aczel, 1988) suggests that autoinjection is not an isolated proposal but part of a broader mathematical tradition that takes self-reference seriously as a structural primitive rather than a pathology. The relationship to fixed point theory (Tarski, 1955) suggests that autoinjection and fixed points are complementary rather than competing concepts. The relationship to Russell's paradox (Russell, 1903) suggests that autoinjection, by restricting self-membership to explicitly constructed sets rather than admitting unrestricted comprehension, avoids the contradictions that motivated the axiom of foundation.

The full development of autoinjective set theory — including a formal axiomatization, consistency proofs, and exploration of the mathematical structures it generates — remains as future work.

9. Conclusion

This paper has introduced autoinjection as a formally distinct class of mathematical mapping in which the range of a function is a subset of its own domain, and the domain contains itself as an element. This class is not present in the classical Bourbaki taxonomy of injection, surjection, and bijection. It is related to but distinct from fixed point theory, non-well-founded set theory, and the self-referential structures studied in the foundations of mathematics. The defining structural property — self-containment of the domain — is proposed not as a pathological condition but as a primitive mathematical property deserving systematic study.

The concept was first introduced in the context of the Special Theory of Everything (Voronin, 2021) as the mathematical foundation for a framework in which every set, concept, and system can be treated as an element of itself. The present paper isolates the purely mathematical content of that concept and presents it independently as a contribution to the foundations of mathematics.

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